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(54) **PROCESS TO PURIFY HELIUM FROM METHANE WITH INTEGRATED NITROGEN REJECTION USING MEMBRANE TECHNOLOGY**

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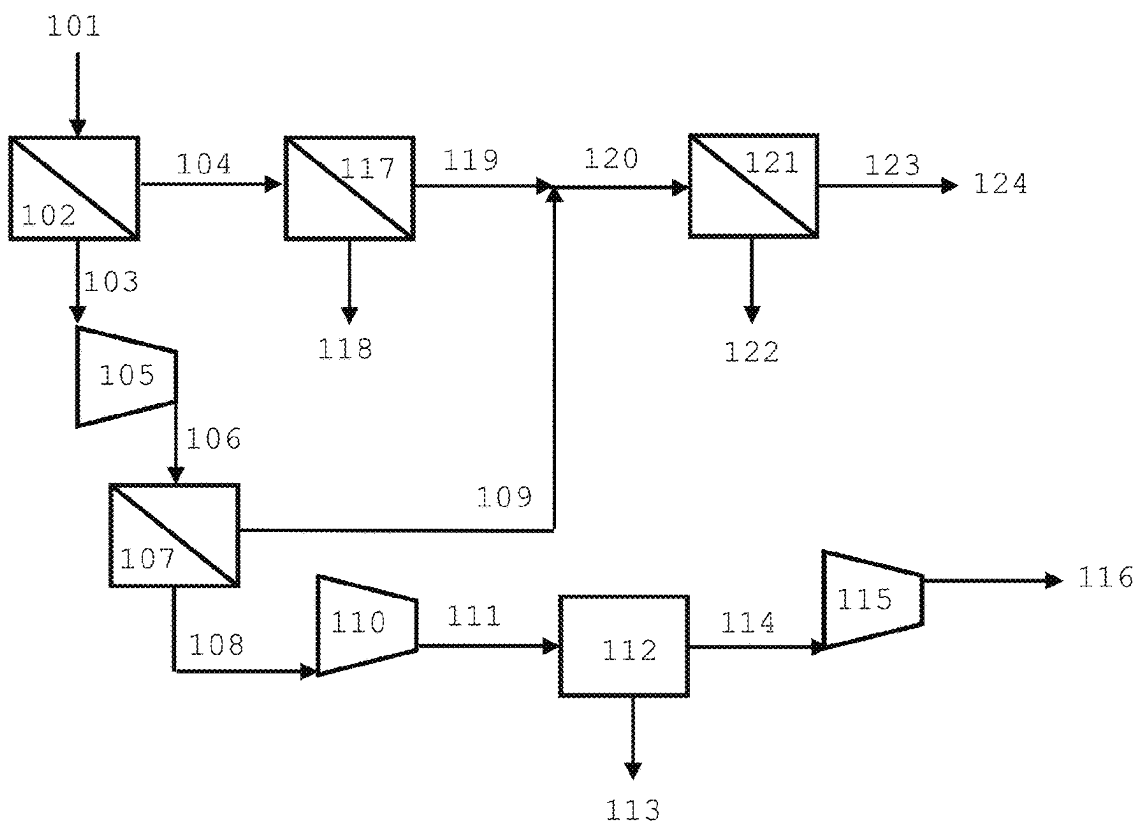
(57) **ABSTRACT**

A process to purify helium from a feed gas stream containing a mixture of at least nitrogen, methane and helium including introducing the feed gas stream into a first helium membrane separation unit, thereby producing a first helium membrane permeate and a first helium membrane residue; introducing at least part of the first residue into a first nitrogen membrane separation unit thereby producing a first nitrogen membrane permeate stream; introducing a stream derived from the first helium membrane permeate into a hydrogen PSA unit thereby producing a helium rich product stream. Wherein a stream derived from the first nitrogen membrane permeate stream exits the system as a fuel gas product stream. Wherein the feed gas stream has a higher heating value, and wherein the first nitrogen membrane permeate stream has a higher heating value at least 5% higher than the higher heating value of the feed gas.



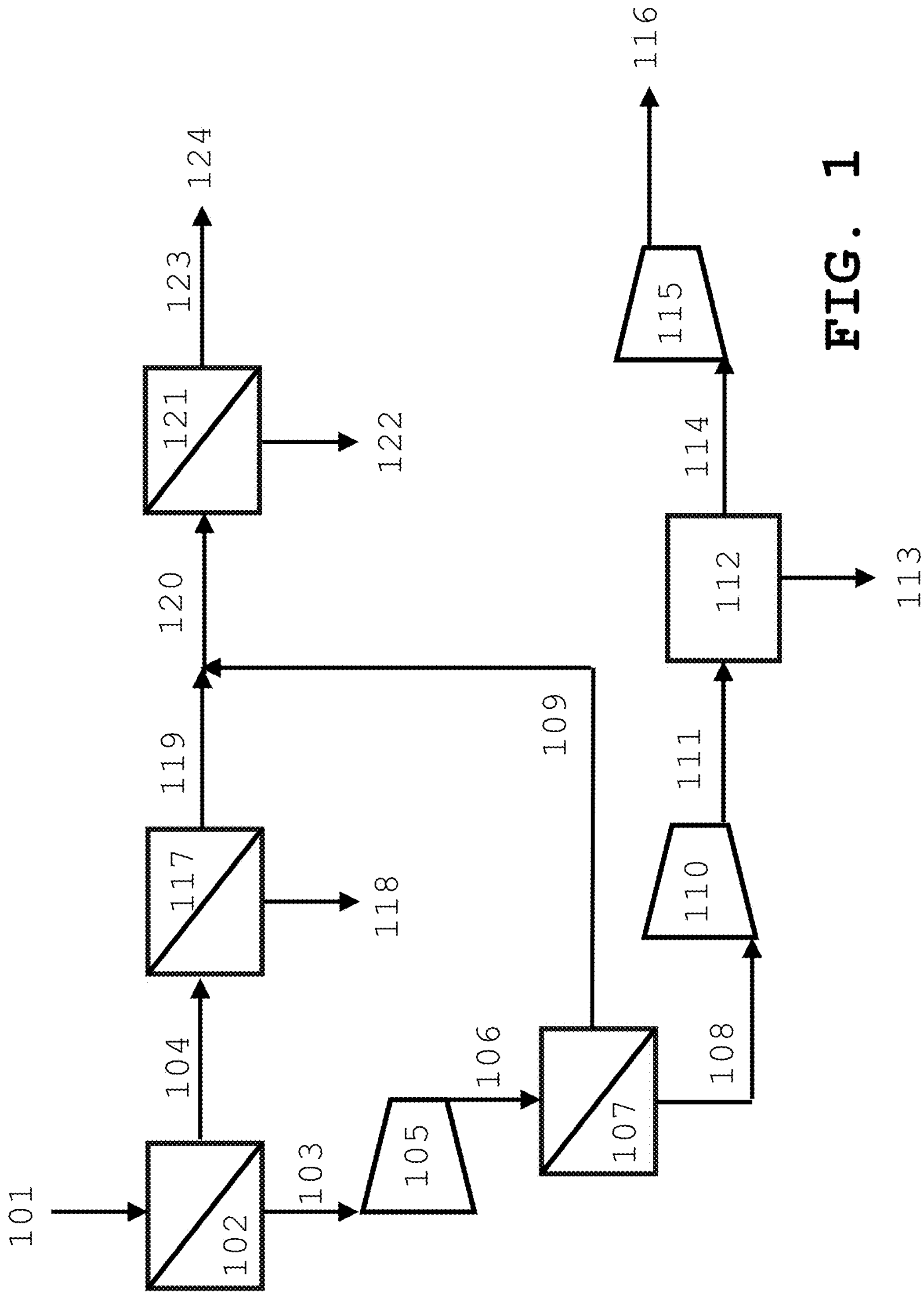

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FIG. 1

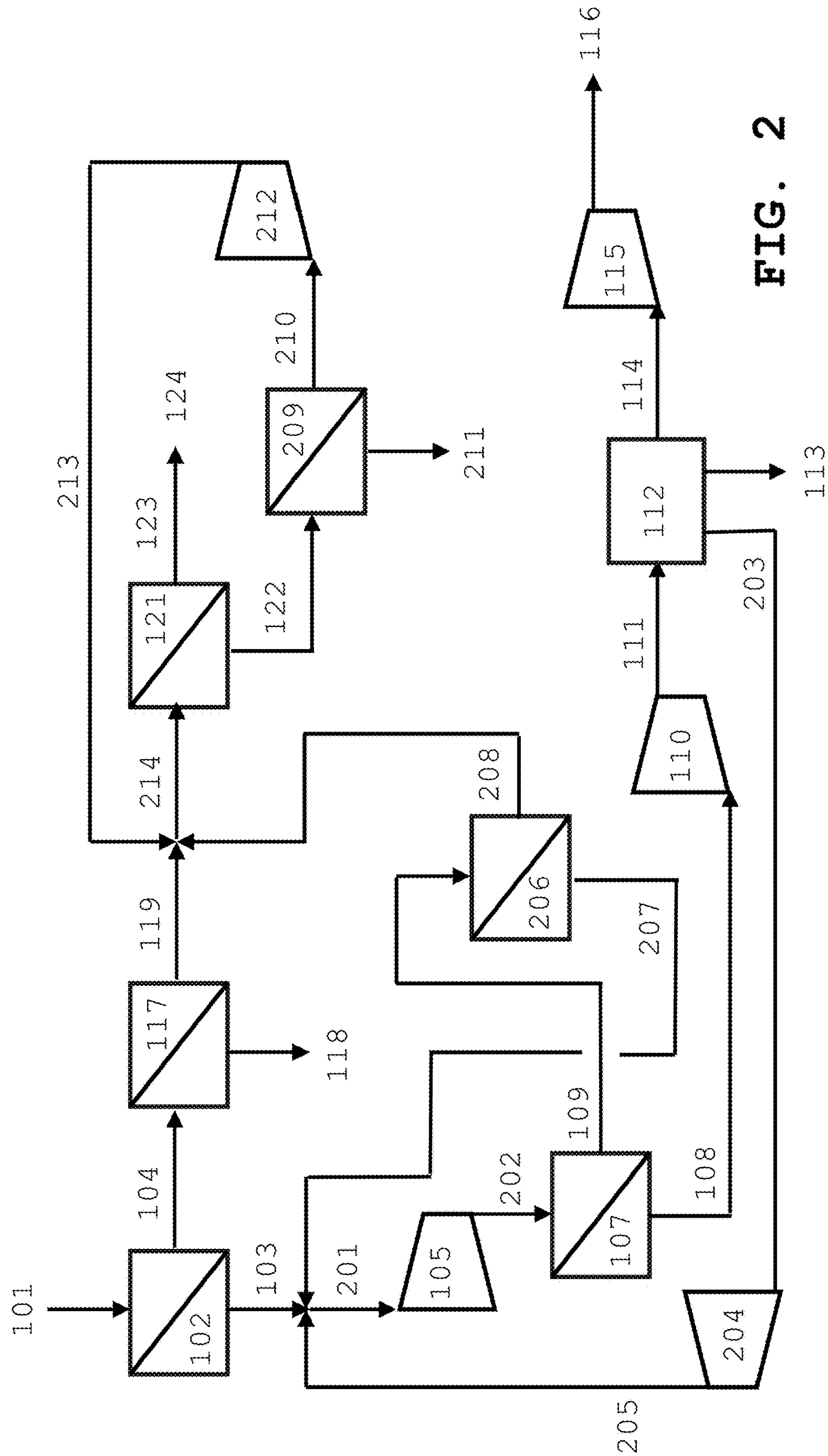
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FIG. 2

PROCESS TO PURIFY HELIUM FROM METHANE WITH INTEGRATED NITROGEN REJECTION USING MEMBRANE TECHNOLOGY

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of priority under 35 U.S.C. § 119 (a) and (b) to U.S. Provisional Patent Application No. 63/559,277, filed Feb. 29, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

[0002] Helium is a valuable molecule that can be found by extracting gas mixtures, like natural gas, from the earth. Since helium is typically a small fraction of the gas mixture, it must be highly purified in order to be used. In some instances, and the waste gas remaining from this process may also contain valuable molecules, like methane, that can be further purified and used.

[0003] While purifying helium from a methane/nitrogen mixture with membrane and pressure swing adsorption (PSA) technology is known in the art, the residue gas is often a mixture of methane and nitrogen in similar proportions as the feed gas. While there are known methods to reject nitrogen from methane, a membrane-based solution is desired as it complements the simplicity of the helium purification process. Proposed is a simple and cost effective process of purifying both helium and natural gas using membrane and adsorption technologies.

SUMMARY

[0004] A process to purify helium from a feed gas stream containing a mixture of at least nitrogen, methane and helium including introducing the feed gas stream into a first helium membrane separation unit, thereby producing a first helium membrane permeate and a first helium membrane residue; introducing at least part of the first residue into a first nitrogen membrane separation unit thereby producing a first nitrogen membrane permeate stream; introducing a stream derived from the first helium membrane permeate into a hydrogen PSA unit thereby producing a helium rich product stream. Wherein a stream derived from the first nitrogen membrane permeate stream exits the system as a fuel gas product stream. Wherein the feed gas stream has a higher heating value, and wherein the first nitrogen membrane permeate stream has a higher heating value at least 5% higher than the higher heating value of the feed gas.

BRIEF DESCRIPTION OF THE FIGURES

[0005] For a further understanding of the nature and objects for the present invention, reference should be made to the following detailed description, taken in conjunction with the accompanying drawings, in which like elements are given the same or analogous reference numbers and wherein:

[0006] FIG. **1** is a schematic representation of a system in accordance with one embodiment of the present invention.

[0007] FIG. **2** is another schematic representation of a system in accordance with one embodiment of the present invention.

ELEMENT NUMBERS

[0008] **101**=feed gas stream

[0009] **102**=first helium membrane separation unit

[0010] **103**=first helium membrane separation permeate stream

[0011] **104**=first helium membrane separation residue stream

[0012] **105**=main compressor

[0013] **106**=compressed first helium membrane separation residue stream

[0014] **107**=second helium membrane separation unit

[0015] **108**=second helium membrane separation permeate stream

[0016] **109**=second helium membrane separation residue stream

[0017] **110**=PSA compressor

[0018] **111**=compressed PSA feed stream

[0019] **112**=helium PSA

[0020] **113**=PSA waste stream

[0021] **114**=purified helium stream

[0022] **115**=helium compressor

[0023] **116**=pressurized helium product stream

[0024] **117**=first nitrogen membrane separation unit

[0025] **118**=first nitrogen membrane separation residue stream

[0026] **119**=first nitrogen membrane separation permeate stream

[0027] **120**=first combined stream

[0028] **121**=second nitrogen membrane separation unit

[0029] **122**=second nitrogen membrane separation residue stream

[0030] **123**=second nitrogen membrane separation permeate stream

[0031] **124**=fuel gas stream

[0032] **201**=second combined stream

[0033] **202**=compressed second combined stream

[0034] **203**=PSA recycle stream

[0035] **204**=first recycle compressor

[0036] **205**=first recycle stream

[0037] **206**=third helium membrane separation unit

[0038] **207**=third helium membrane separation permeate stream

[0039] **208**=third helium membrane separation residue stream

[0040] **209**=third nitrogen membrane separation unit

[0041] **210**=third nitrogen membrane separation permeate stream

[0042] **211**=third nitrogen membrane separation residue stream

[0043] **212**=second recycle compressor

[0044] **213**=second recycle stream

[0045] **214**=third combined stream

DESCRIPTION OF PREFERRED EMBODIMENTS

[0046] Illustrative embodiments of the invention are described below. While the invention is susceptible to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the drawings and are herein described in detail. It should be understood, however, that the description herein of specific embodiments is not intended to limit the invention to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention as defined by the appended claims.

[0047] It will of course be appreciated that in the development of any such actual embodiment, numerous implementation-specific decisions must be made to achieve the developer's specific goals, such as compliance with system-related and business-related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine undertaking for those of ordinary skill in the art having the benefit of this disclosure.

[0048] The following description explicitly refers to helium as the target gas, but the skilled artisan will also recognize that this same process may be utilized to separate hydrogen from a feed gas stream containing a mixture of at least nitrogen, methane and hydrogen. One of ordinary skill in the art will recognize that modifying the following process will involve the selection of hydrogen permeable membranes and a PSA that is designed to purify hydrogen. But with little experimentation, the skilled artisan will be able to modify this process to purify hydrogen.

[0049] As used herein, the term "approximately the same" is defined as being within 10%, preferably 5%, and more preferably 1% of the same measured value.

[0050] As used herein, the term "Helium rich" is defined as having a Helium concentration greater than 97% by volume, preferably greater than 98% by volume.

[0051] As used herein, the term "ultra-high purity Helium" is defined as having a helium concentration greater than 99.9% by volume, preferably greater than 99.99% by volume, more preferably greater than 99.999% by volume.

[0052] As used herein, the term "helium membrane separation unit" is defined as a device that utilizes a semipermeable barrier (membrane) that is selective for helium, and thus is used to separate helium from a mixture of gases.

[0053] As used herein, the term "helium pressure swing adsorption unit" is defined as a device that utilizes a adsorbent beds that are selective for helium, and thus is used to separate helium from a mixture of gases.

[0054] The present invention may be used to retrofit an existing process that utilizes membrane separation (and optionally additional helium pressure swing adsorption (PSA)) for helium purification from methane, nitrogen, or other slow gases. A key innovative element of this process is the addition of a nitrogen rejection unit (NRU) using membranes to further purify the helium process off-gas by removing any residual nitrogen, thus resulting in a usable fuel gas.

[0055] Turning to FIG. **1**, one embodiment of the present invention is presented. Feed gas stream **101** is introduced into first helium membrane separation unit **102**, thereby producing first helium membrane separation permeate stream **103** and first helium membrane separation residue stream **104**. Feed gas stream **101** is comprised of, at least, nitrogen, methane, and helium. Feed gas stream **101** has a first Higher Heating Value (HHV). First helium membrane separation residue stream **104** has a first pressure P**1**. First helium membrane separation permeate stream **103** is introduced into main compressor **105**, thereby producing compressed first helium membrane separation permeate stream **106**. Compressed first helium membrane separation permeate stream **106** is introduced into second helium membrane separation unit **107**, thereby producing second helium membrane separation residue stream **109** and second helium membrane separation permeate stream **108**. Second helium membrane separation residue stream **109** has a second pressure P**2**.

[0056] Second helium membrane separation permeate stream **108** is introduced into PSA compressor **110**, thereby producing compressed PSA feed stream **111**. Compressed PSA feed stream **111** is introduced into helium PSA **112**, thereby producing PSA waste stream **113** and purified helium stream **114**. Purified helium stream **114** comprises a Ultra-High purity Helium stream Purified helium stream **114** is introduced into helium compressor **115**, thereby producing pressurized helium product stream **116**.

[0057] First helium membrane separation residue stream **104** is introduced into first nitrogen membrane separation unit **117**, thereby producing first nitrogen membrane separation residue stream **118** and first nitrogen membrane separation permeate stream **119**. First nitrogen membrane separation permeate stream **119** has a third pressure P**3**. First nitrogen membrane separation permeate stream **119** has a second higher heating value. In one embodiment, the second higher heating value is at least 5% higher than the first higher heating value. In one embodiment of the present invention, P**2** is as close as possible to P**3**. First nitrogen membrane separation residue stream **118** is generally considered to be a waste stream. First nitrogen membrane separation permeate stream **119** is combined with second helium membrane separation residue stream **109** to produce first combined stream **120**. First combined stream **120** is introduced into second nitrogen membrane separation unit **121**, thereby producing second nitrogen membrane separation residue stream **122** and second nitrogen membrane separation permeate stream **123**. Second nitrogen membrane separation permeate stream **123** has a fourth pressure P**4**. Second nitrogen membrane separation residue stream **122** is generally considered to be a waste stream. Second nitrogen membrane separation permeate stream **123** leaves the system as fuel gas stream **124**.

[0058] Turning to FIG. **2**, another embodiment of the present invention is presented. Feed gas **101** is introduced into first helium membrane separation unit **102**, thereby producing first helium membrane separation permeate stream **103** and first helium membrane separation residue stream **104**. Feed gas stream **101** is comprised of, at least, nitrogen, methane, and helium. Feed gas stream **101** has a first Higher Heating Value (HHV). First helium membrane separation permeate stream **103** is combined with first recycle stream **205** and third helium membrane separation permeate stream **207** thereby producing second combined stream **201**. Second combined stream **201** is introduced into main compressor **105**, thereby producing compressed second combined stream **202**. Compressed second combined stream **202** is introduced into second helium membrane separation unit **107**, thereby producing second helium membrane separation permeate stream **108** and second helium membrane separation residue stream **109**. Second helium membrane separation permeate stream **108** comprises a Helium rich stream. Second helium membrane separation permeate stream **108** is introduced into third helium membrane separation unit **206**, thereby producing third helium membrane separation permeate stream **207** and third helium membrane separation residue stream **208**. Second helium membrane separation residue stream **109** is introduced into

PSA compressor **110**, thereby producing compressed PSA feed stream **111**. Compressed PSA feed stream **111** is introduced into helium PSA **112**, thereby producing PSA waste stream **113**, PSA recycle stream **203**, and purified helium stream **114**. Purified helium stream **114** comprises a Ultra-High purity Helium stream. Purified helium stream **114** is introduced into helium compressor **115**, thereby producing pressurized helium product stream **116**. PSA recycle stream **203** is introduced into first recycle compressor **204**, thereby producing first recycle stream **205**.

[0059] First helium membrane separation residue stream **104** is introduced into first nitrogen membrane separation unit **117**, thereby producing first nitrogen membrane separation residue stream **118** and first nitrogen membrane separation permeate stream **119**. First nitrogen membrane separation permeate stream **119** has a second higher heating value. In one embodiment, the second higher heating value is at least 5% higher than the first higher heating value. First nitrogen membrane separation permeate stream **119** has a fifth pressure P**5**. In one embodiment of the present invention, P**2**, P**4**, and P**5** are as close as possible to one another. First nitrogen membrane separation residue stream **118** is generally considered to be a waste stream. First nitrogen membrane separation permeate stream **119** is combined with third helium membrane separation residue stream **208** and second compressed recycle stream **213** to produce third combined stream **214**. Third combined stream **214** is introduced into second nitrogen membrane separation unit **121**, thereby producing second nitrogen membrane separation residue stream **122** and second nitrogen membrane separation permeate stream **123**. Second nitrogen membrane separation residue stream **122** is generally considered to be a waste stream. Second nitrogen membrane separation permeate stream **123** leaves the system as fuel gas stream **124**.

[0060] Second nitrogen membrane separation residue stream **122** is introduced into third nitrogen membrane separation unit **209**, thereby producing third nitrogen membrane separation residue stream **211** and third nitrogen membrane separation permeate stream **210**. Third nitrogen membrane separation residue stream **211** is generally consider to be a waste stream. Third nitrogen membrane separation permeate stream **210** is introduced into second recycle compressor **212**, thereby producing second compressed recycle stream **213**.

[0061] It will be understood that many additional changes in the details, materials, steps and arrangement of parts, which have been herein described in order to explain the nature of the invention, may be made by those skilled in the art within the principle and scope of the invention as expressed in the appended claims. Thus, the present invention is not intended to be limited to the specific embodiments in the examples given above.

What is claimed is:

1: A process to purify helium from a feed gas stream **101** containing a mixture of at least nitrogen, methane and helium comprising:

introducing the feed gas stream **101** into a first helium membrane separation unit **102**, thereby producing a first helium membrane permeate **103** and a first helium membrane residue **104** at a first pressure P**1**;

introducing at least part of the first residue **104** in a first nitrogen membrane separation unit **117** thereby producing a first nitrogen membrane permeate stream **119** at a third pressure P**3** and a first nitrogen membrane residue stream **118**;

introducing a stream derived from the first helium membrane permeate **103** into a hydrogen PSA unit **112** thereby producing a helium rich product stream **116**;

wherein a stream derived from the first nitrogen membrane permeate stream **119** exits the system as a fuel gas product stream **124**;

wherein the feed gas stream **101** has a first higher heating value, and

wherein the first nitrogen membrane permeate stream **119** has a second higher heating value,

wherein the second higher heating value is at least 5% higher than the first higher heating value.

2: The process according to claim **1**, further comprising:

compressing the first helium membrane permeate **103**, thereby producing a compressed first helium membrane permeate stream **106**, and introducing the compressed first helium membrane permeate stream **106** into a second helium membrane separation unit **107**, thereby obtaining a second helium membrane residue stream **109** at a second pressure P**2**, and a second helium membrane permeate stream **108** further enriched in helium;

wherein, at least part of the first nitrogen membrane permeate stream **119** and at least part of a stream derived from the second helium membrane residue stream **109** are combined **112** and introduced in a second nitrogen membrane separation unit **121** thereby obtaining a second nitrogen membrane separation permeate stream **123** and a second nitrogen membrane separation residue stream **122**;

wherein the fuel gas product stream **124** is derived from the second nitrogen membrane separation permeate stream **123**, and

wherein the first nitrogen membrane permeate stream **119** has a second higher heating value,

wherein the second higher heating value is at least 5% higher than the first higher heating value.

3: The process according to claim **2**,

wherein the second nitrogen membrane separation permeate stream **123** has a fourth pressure P**4**;

wherein the first nitrogen membrane residue stream **118** has a fifth pressure P**5**, and

wherein P**2**, P**4**, and P**5** are approximately the same.

4: A process to purify helium from a feed gas stream **101** containing a mixture of at least nitrogen, methane and helium comprising:

introducing the feed gas stream **101** into a first helium membrane separation unit **102**, thereby producing a first helium membrane permeate **103** and a first helium membrane residue **104**;

introducing at least part of the first residue **104** in a first nitrogen membrane separation unit **117** thereby producing a first nitrogen membrane permeate stream **119** and a first nitrogen membrane residue stream **118**;

introducing at least part of the first nitrogen membrane permeate stream **119** into a second nitrogen membrane separation unit **121**, thereby producing a second nitrogen membrane permeate stream **123** and a second nitrogen membrane residue stream **122**;

introducing at least part of the second nitrogen membrane residue stream **122** into a third nitrogen membrane separation unit **209**, thereby producing a third nitrogen membrane permeate stream **210** and a third nitrogen membrane residue stream **211**;

and

introducing a stream derived from the second helium membrane permeate stream **109** into a hydrogen PSA unit **112** thereby producing a helium rich product stream **116**;

wherein a stream derived from the second nitrogen membrane permeate stream **123** exits the system as a fuel gas product stream **124**.

5: The process according to claim **4**, further comprising a recycle compressor,

wherein at least part of the third nitrogen membrane permeate stream **210** is recycled and combined with at least part of the first nitrogen membrane permeate stream **119**.

6: The process according to claim **4**, further comprising a recycle compressor,

wherein at least part of the third nitrogen membrane permeate stream **210** is recycled and combined with at least part of the first nitrogen membrane permeate stream **119** and at least part of the third helium membrane residue stream **208**.

7: The process of claim **4**, further comprising:

introducing at least part of the first helium membrane permeate **103** into a second helium membrane separation unit **107**, thereby producing a second helium membrane residue stream **109** and a second helium membrane permeate stream **108**; and

introducing at least part of the second helium membrane permeate stream **108** into a third helium membrane separation unit **206**, thereby producing a third helium membrane permeate stream **207** and a third helium membrane residue stream **208**.

8: A process to purify hydrogen from a feed gas stream **101** containing a mixture of at least nitrogen, methane and hydrogen comprising:

introducing the feed gas stream **101** into a first hydrogen membrane separation unit **102**, thereby producing a first hydrogen membrane permeate **103** and a first hydrogen membrane residue **104** at a first pressure P**1**;

introducing at least part of the first residue **104** in a first nitrogen membrane separation unit **117** thereby producing a first nitrogen membrane permeate stream **119** at a third pressure P**3** and a first nitrogen membrane residue stream **118**;

introducing a stream derived from the first hydrogen membrane permeate **103** into a hydrogen PSA unit **112** thereby producing a hydrogen rich product stream **116**;

wherein a stream derived from the first nitrogen membrane permeate stream **119** exits the system as a fuel gas product stream **124**;

wherein the feed gas stream **101** has a first higher heating value, and

wherein the first nitrogen membrane permeate stream **119** has a second higher heating value,

wherein the second higher heating value is at least 5% higher than the first higher heating value.

9: The process according to claim **8**, further comprising:

compressing the first hydrogen membrane permeate **103**, thereby producing a compressed first hydrogen membrane permeate stream **106**, and introducing the compressed first hydrogen membrane permeate stream **106** into a second hydrogen membrane separation unit **107**, thereby obtaining a second hydrogen membrane residue stream **109** at a second pressure P**2**, and a second hydrogen membrane permeate stream **108** further enriched in hydrogen;

wherein, at least part of the first nitrogen membrane permeate stream **119** and at least part of a stream derived from the second hydrogen membrane residue stream **109** are combined **112** and introduced in a second nitrogen membrane separation unit **121** thereby obtaining a second nitrogen membrane separation permeate stream **123** and a second nitrogen membrane separation residue stream **122**;

wherein the fuel gas product stream **124** is derived from the second nitrogen membrane separation permeate stream **123**, and

wherein the first nitrogen membrane permeate stream **119** has a second higher heating value,

wherein the second higher heating value is at least 5% higher than the first higher heating value.

10: The process according to claim **9**,

wherein the second nitrogen membrane separation permeate stream **123** has a fourth pressure P**4**;

wherein the first nitrogen membrane residue stream **118** has a fifth pressure P**5**, and

wherein P**2**, P**4**, and P**5** are approximately the same.

11: A process to purify hydrogen from a feed gas stream **101** containing a mixture of at least nitrogen, methane and hydrogen comprising:

introducing the feed gas stream **101** into a first hydrogen membrane separation unit **102**, thereby producing a first hydrogen membrane permeate **103** and a first hydrogen membrane residue **104**;

introducing at least part of the first residue **104** in a first nitrogen membrane separation unit **117** thereby producing a first nitrogen membrane permeate stream **119** and a first nitrogen membrane residue stream **118**;

introducing at least part of the first nitrogen membrane permeate stream **119** into a second nitrogen membrane separation unit **121**, thereby producing a second nitrogen membrane permeate stream **123** and a second nitrogen membrane residue stream **122**;

introducing at least part of the second nitrogen membrane residue stream **122** into a third nitrogen membrane separation unit **209**, thereby producing a third nitrogen membrane permeate stream **210** and a third nitrogen membrane residue stream **211**;

and

introducing a stream derived from the second hydrogen membrane permeate stream **109** into a hydrogen PSA unit **112** thereby producing a hydrogen rich product stream **116**;

wherein a stream derived from the second nitrogen membrane permeate stream **123** exits the system as a fuel gas product stream **124**.

12: The process according to claim **11**, further comprising a recycle compressor,

wherein at least part of the third nitrogen membrane permeate stream **210** is recycled and combined with at least part of the first nitrogen membrane permeate stream **119**.

13: The process according to claim **11**, further comprising a recycle compressor,

wherein at least part of the third nitrogen membrane permeate stream **210** is recycled and combined with at least part of the first nitrogen membrane permeate stream **119** and at least part of the third hydrogen membrane residue stream **208**.

14: The process of claim **11**, further comprising:

introducing at least part of the first hydrogen membrane permeate **103** into a second hydrogen membrane separation unit **107**, thereby producing a second hydrogen membrane residue stream **109** and a second hydrogen membrane permeate stream **108**; and

introducing at least part of the second hydrogen membrane permeate stream **108** into a third hydrogen membrane separation unit **206**, thereby producing a third hydrogen membrane permeate stream **207** and a third hydrogen membrane residue stream **208**.

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